

DDL,DML,DCL & TCL Commands:

Program:

```
1      -- Step 1: Create Database
2  •    CREATE DATABASE bank_management_db;
3  •    USE bank_management_db;
4      -- Step 2: Create Bank Transactions Table
5  •    CREATE TABLE bank_transactions(
6          txn_id INT PRIMARY KEY,
7          customer_name VARCHAR(50),
8          branch_name VARCHAR(50),
9          transaction_type VARCHAR(20),
10         amount DECIMAL(10,2),
11         transaction_date DATE
12     );
13     -- Step 3: Add Account Number Column
14  •    ALTER TABLE bank_transactions
```

Output:

#	Time	Action	Message	Duration / Fetch
✓ 1	14:05:06	CREATE DATABASE bank_management_db	1 row(s) affected	0.000 sec
✓ 2	14:05:07	USE bank_management_db	0 row(s) affected	0.000 sec
✓ 3	14:05:10	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, ...	0 row(s) affected	0.031 sec
✓ 4	14:05:13	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec

Program:

```
15      ADD account_no VARCHAR(20);
16      -- Step 4: Modify Customer Name Column
17      • ALTER TABLE bank_transactions
18      MODIFY customer_name VARCHAR(100);
19      -- Step 5: Rename Transaction Table
20      • RENAME TABLE bank_transactions
21      TO customer_transactions;
22      -- Step 6: Truncate Transaction Table
23      • TRUNCATE TABLE customer_transactions;
24      -- Step 7: Create Backup Table
```

Output:

Output					
Action Output					
#	Time	Action	Message	Duration / Fetch	
✓	4 14:05:13	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec	
✗	5 14:06:45	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	Error Code: 1060. Duplicate column name 'account_no'	0.016 sec	
✓	6 14:07:00	ALTER TABLE bank_transactions MODIFY customer_name VARC...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec	
✓	7 14:07:04	ALTER TABLE bank_transactions MODIFY customer_name VARC...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec	
✓	8 14:07:13	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.015 sec	
✓	9 14:07:19	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.031 sec	

Program:

```
24 -- Step 7: Create Backup Table
25 • CREATE TABLE bank_backup(
26     txn_id INT,
27     customer_name VARCHAR(100)
28 );
29 -- Step 8: Drop Backup Table
30 • DROP TABLE bank_backup;
31 -- Step 9: Insert Sample Data
32 • INSERT INTO customer_transactions
33 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
34 VALUES
35 (101, 'Tejaswini', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
36 (102, 'Pranitha', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
37 (103, 'Ramu', 'Hyderabad', 'Deposit', 12000, '2024-01-08'),
38 (104, 'Pavani', 'Vizag', 'Deposit', 8000, '2024-01-10'),
39 (105, 'Nikila', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
40 (106, 'Manu', 'Vizag', 'Deposit', 15000, '2024-01-12'),
41 (107, 'Sahas', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
42 (108, 'Santhvin', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
43 (109, 'Ravi', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
44 (110, 'Lahari', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
45 -- Step 10: Insert New Transaction
```

Output:

✓	10	14:08:45	CREATE TABLE bank_backup(txn_id INT, customer_name V...	0 row(s) affected	0.015 sec
✓	11	14:08:49	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
✗	12	14:08:55	INSERT INTO customer_transactions (txn_id, customer_name, bran...	Error Code: 1136. Column count doesn't match value count at row 3	0.016 sec
✓	13	14:10:32	INSERT INTO customer_transactions (txn_id, customer_name, bran...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

Program:

```
45 -- Step 10: Insert New Transaction
46 • INSERT INTO customer_transactions
47 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
48 VALUES
49 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
50 -- Step 11: Update Transaction Amount
51 • UPDATE customer_transactions
52 SET amount = 5000
53 WHERE txn_id = 105;
54 -- Step 12: Delete Transaction
55 • DELETE FROM customer_transactions
56 WHERE txn_id = 111;
```

Output:

✓	14	14:11:54	INSERT INTO customer_transactions (txn_id, customer_name, bran...	1 row(s) affected	0.016 sec
✓	15	14:11:59	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
✓	16	14:12:12	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.016 sec

Program:

```
SELECT *
FROM customer_transactions;
```

Output:

Result Grid						
txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Tejaswini	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Pranitha	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Ramu	Hyderabad	Deposit	12000.00	2024-01-08	NULL
104	Pavani	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Nikila	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Sahas	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Ravi	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL
111	Venu	Vizag	Deposit	7000.00	2024-01-18	NULL

Output				
#	Time	Action	Message	Duration / Fetch
✗	12	14:08:55	INSERT INTO customer_transactions (txn_id, customer_name, bra... Error Code: 1136. Column count doesn't match value count at row 3	0.016 sec
✓	13	14:10:32	INSERT INTO customer_transactions (txn_id, customer_name, bra... 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
✓	14	14:11:54	INSERT INTO customer_transactions (txn_id, customer_name, bra... 1 row(s) affected	0.016 sec
✓	15	14:11:59	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
✓	16	14:12:12	DELETE FROM customer_transactions WHERE txn_id = 111 1 row(s) affected	0.016 sec
✓	17	14:12:57	SELECT * FROM customer_transactions LIMIT 0, 1000 10 row(s) returned	0.000 sec / 0.000 sec

Program:

```
60  -- Step 14: Display Deposit Transactions
61  •  SELECT *
62     FROM customer_transactions
63     WHERE transaction_type = 'Deposit';
```

Output:

Result Grid							
Filter Rows:							
Edit: Export/Import: Wrap Cell Cont							
txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no	
101	Tejaswini	Hyderabad	Deposit	5000.00	2024-01-05	NULL	
103	Ramu	Hyderabad	Deposit	12000.00	2024-01-08	NULL	
104	Pavani	Vizag	Deposit	8000.00	2024-01-10	NULL	
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL	
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL	
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL	
*	NULL	NULL	NULL	NULL	NULL	NULL	

Result Grid

Form Editor

Field Types

Query Stats

customer_transactions 2 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 13	14:10:32	INSERT INTO customer_transactions (txn_id, customer_name, bra...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
✓ 14	14:11:54	INSERT INTO customer_transactions (txn_id, customer_name, bra...	1 row(s) affected	0.016 sec
✓ 15	14:11:59	UPDATE customer_transactions SET amount = 5000 WHERE txn...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
✓ 16	14:12:12	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.016 sec
✓ 17	14:12:57	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 18	14:13:41	SELECT * FROM customer_transactions WHERE transaction_type...	6 row(s) returned	0.000 sec / 0.000 sec

Program:

```
64 -- Step 15: Sort Transactions by Amount
65 • SELECT *
66 FROM customer_transactions
67 ORDER BY amount DESC;
```

Output:

Result Grid							
Filter Rows:							
Edit: Export/Import: Wrap Cell Cont							
txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no	
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL	
103	Ramu	Hyderabad	Deposit	12000.00	2024-01-08	NULL	
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL	
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL	
104	Pavani	Vizag	Deposit	8000.00	2024-01-10	NULL	
101	Tejaswini	Hyderabad	Deposit	5000.00	2024-01-05	NULL	
105	Nikla	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL	
109	Ravi	Vizag	Withdrawal	4000.00	2024-01-15	NULL	
102	Pranitha	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL	
107	Sahas	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL	
*	NULL	NULL	NULL	NULL	NULL	NULL	

customer_transactions 3 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 14	14:11:54	INSERT INTO customer_transactions (txn_id, customer_name, bra...	1 row(s) affected	0.016 sec
✓ 15	14:11:59	UPDATE customer_transactions SET amount = 5000 WHERE txn...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
✓ 16	14:12:12	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected	0.016 sec
✓ 17	14:12:57	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 18	14:13:41	SELECT * FROM customer_transactions WHERE transaction_type...	6 row(s) returned	0.000 sec / 0.000 sec
✓ 19	14:14:13	SELECT * FROM customer_transactions ORDER BY amount DES...	10 row(s) returned	0.000 sec / 0.000 sec

Program:

```
68      -- Create Auditor1 User
69  ●   CREATE USER 'Auditor1'@'localhost'
70      IDENTIFIED BY '1234';
71
72      -- Create BranchManager User
73  ●   CREATE USER 'BranchManager'@'localhost'
74      IDENTIFIED BY '1234';
75      -- Step 16: Grant SELECT Permission
76  ●   GRANT SELECT
77      ON customer_transactions
78      TO 'Auditor1'@'localhost';
79      -- Step 17: Grant All Privileges
80  ●   GRANT ALL
81      ON customer_transactions
82      TO 'BranchManager'@'localhost';
83      -- Step 18: Revoke SELECT Permission
84  ●   REVOKE SELECT
85      ON customer_transactions
86      FROM 'Auditor1'@'localhost';
```

Output:

Output					
Action Output					
#	Time	Action	Message	Duration / Fetch	
✓ 19	14:14:13	SELECT * FROM customer_transactions ORDER BY amount DES...	10 row(s) returned	0.000 sec / 0.000 sec	
✓ 20	14:15:15	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected	0.046 sec	
✓ 21	14:15:22	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '12...	0 row(s) affected	0.016 sec	
✓ 22	14:15:28	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localh...	0 row(s) affected	0.016 sec	
✓ 23	14:15:32	GRANT ALL ON customer_transactions TO 'BranchManager'@'loc...	0 row(s) affected	0.015 sec	
✓ 24	14:15:37	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@1...	0 row(s) affected	0.000 sec	

Program:

```
REVOKE ALL PRIVILEGES
ON customer_transactions
FROM 'BranchManager'@'localhost';
-- Step 20: Commit Transaction

UPDATE customer_transactions
SET amount = 2000
WHERE txn_id = 102;

-- Step 21: Create Savepoint
START TRANSACTION;

SAVEPOINT Before_Update;

-- Step 22: Rollback to Savepoint
UPDATE customer_transactions
SET amount = 99999
WHERE txn_id = 102;

ROLLBACK TO SAVEPOINT Before_Update;
-- Reset Transaction State
ROLLBACK;
```

Output:

Output				Context help	Snippets
Action Output					
#	Time	Action	Message	Duration / Fetch	
✓ 25	14:16:59	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'Bra...	0 row(s) affected	0.016 sec	
✓ 26	14:17:06	UPDATE customer_transactions SET amount = 2000 WHERE txn...	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec	
✓ 27	14:17:22	START TRANSACTION	0 row(s) affected	0.000 sec	
✓ 28	14:17:24	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec	
✓ 29	14:17:27	UPDATE customer_transactions SET amount = 99999 WHERE tx...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec	
✓ 30	14:17:30	ROLLBACK TO SAVEPOINT Before Update	0 row(s) affected	0.016 sec	
✓ 31	14:17:33	ROLLBACK	0 row(s) affected	0.000 sec	

Program:

```
SET autocommit = 1;
-- Step 23: Complete TCL Example
START TRANSACTION;

UPDATE customer_transactions
SET amount = 7000
WHERE txn_id = 101;

SAVEPOINT SP1;

UPDATE customer_transactions
SET amount = 9000
WHERE txn_id = 102;

ROLLBACK TO SAVEPOINT SP1;

COMMIT;
```

Output:

Output					Context Help	Snippets
Action Output						
#	Time	Action	Message	Duration / Fetch		
✓ 32	14:20:07	SET autocommit = 1	0 row(s) affected	0.000 sec		
✓ 33	14:20:11	START TRANSACTION	0 row(s) affected	0.000 sec		
✓ 34	14:20:14	UPDATE customer_transactions SET amount = 7000 WHERE txn...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec		
✓ 35	14:20:16	SAVEPOINT SP1	0 row(s) affected	0.000 sec		
✓ 36	14:20:19	UPDATE customer_transactions SET amount = 9000 WHERE txn...	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec		
✓ 37	14:20:21	ROLLBACK TO SAVEPOINT SP1	0 row(s) affected	0.000 sec		
✓ 38	14:20:26	COMMIT	0 row(s) affected	0.000 sec		